

Module 2

Motherboards: Motherboard controllers and system resources, I/O system bus, on-board I/O devices, Chipsets, ROM BIOS, ROM POST, CMOS setup, Motherboard physical form factors. Power supply, cooling and protection: The power supply, Ventilation and cooling protection, Power protection and backup.

Motherboards

The motherboard is the main circuit board of your computer and is also known as the mainboard or logic board. Attached to the motherboard, the CPU, ROM, memory RAM expansion slots, PCI slots, and USB ports. It also includes controllers for devices like the hard drive, DVD drive, keyboard, and mouse. Basically, the motherboard is what makes everything in your computer work together. Each motherboard has a collection of chips and controllers known as the chipset. A motherboard is the main printed circuit board (PCB) in a computer. The most common motherboard design in desktop computers today is ATX, an Intel improvement on the AT design by IBM.

Motherboard controllers and system resources

System resources are resources that are electrically used by hardware, allocated by firmware, and used by software for each device. A system resource is any usable part of a computer that can be controlled and assigned by the [operating system](#) so all of the [hardware](#) and [software](#) on the computer can work together as designed. In a computer system, system resources are the components that provide its inherent capabilities and contribute to its overall performance. System memory, cache memory, hard disk space, IRQs and DMA channels are examples.

I/O system bus

The input/output bus or io bus is the pathway used for input and output devices to communicate with the computer processor. The I/O system bus provides a communication path for the data and control signals moving between the major components of the computer system. The system bus works by combining the functions of the three main buses: namely, the data, address and control buses. Each of the three buses has its separate characteristics and responsibilities.

Some of the I/O System Bus are

- Industry Standard Architecture
- Micro Channel Architecture
- Enhanced Industry Standard Architecture
- VESA Local Bus
- Peripheral Component Interconnect
- Accelerated Graphics Port

- PCI-X

Industry Standard Architecture (ISA)

- ISA was the first of the open standards released by IBM in an effort to develop and implement a standard expansion bus.
- ISA (Industry Standard Architecture) is a standard bus (computer interconnection) architecture that was associated with the IBM AT motherboard.
- ISA is a type of bus used in PCs for adding expansion cards. For example, an ISA slot may be used to add a video card, a network card, or an extra serial port.
- ISA was originally an 8-bit computer bus that was later expanded to a 16-bit bus in 1984.
- The new ISA bus was flexible in that it could connect multiple devices. It supported 16-bit peripheral devices.

- **Micro Channel Architecture(MCA)**

- It is an expansion bus created by IBM that was used in the company's PS/2 desktop computers.
- The pin connections in MCA are smaller than other bus interfaces. For this and other reasons, MCA does not support other bus architectures. Although MCA offers a number of improvements over other bus architectures.
- Plug and Play: MCA automatically configured adapter cards, so there was no need to fiddle with jumpers.
- The MCA bus supported bus mastering adapters for greater efficiency, including proper bus arbitration.
- A direct memory access (DMA) procedure that supports multiple DMA channels. Additionally, this procedure allows a device to transfer data in bursts.
- Support for audio signal transfer.
- Support for both synchronous and asynchronous data transfer.

Enhanced Industry Standard Architecture (EISA)

- Extended Industry Standard Architecture (EISA) is a bus architecture that extends the Industry Standard Architecture (ISA) from 16 bits to 32 bits.
- EISA extended the advanced technology (AT) bus architecture and facilitated bus sharing between multiple central processing units (CPU).
- The EISA bus is compatible with older ISA buses with 8-bit or 16-bit data paths.
- EISA was primarily used for high-end servers requiring heavy bandwidth.
- providing more signals and enhanced performance.

VESA Local Bus (Video Electronics Standards Association)

- VL-bus is a hardware interface on the computer's motherboard that is attached to an expansion slot.
- By connecting a video expansion card to the VLB, you can add extra graphics capabilities to your computer.
- The interface supports 32-bit data flow at up to 50 MHz.
- It was invented to help standardize PC's video specifications.
- The design specification provides two other performance-boosting features: *burst mode* and *bus mastering*

Peripheral Component Interconnect (PCI)

- Peripheral Component Interconnect is a common connection interface for attaching computer peripherals to the motherboard.
- PCI slots are utilized to install sound cards, Ethernet and remote cards.
- Adding PCI cards was an easy way to upgrade a computer, since you could add a better video card, faster wired or wireless networking, or add new ports, like USB 2.0.
- PCI didn't require jumpers or dip switches, as ISA did.
- PCI does not support pipeline.
- Peripheral component interconnect (PCI) was developed by Intel as a processor-independent, high-speed replacement for ISA.
- PCI can coexist with other buses, such as ISA, on the same motherboard. Many PCs have both ISA and PCI slots.
- The PCI bus can operate in either a synchronous or asynchronous mode.
- PCI devices are capable of bus mastering.

Accelerated Graphics Port (AGP)

- AGP is an advanced port designed for video cards and 3D accelerators. Developed by Intel.
- AGP introduces a dedicated point-to-point channel that allows the graphics controller direct access to the system memory.
- The AGP channel is 32-bits wide and runs at 66 MHz.
- Much greater than the PCI bandwidth.
- It also allows 3-D textures to be stored in main memory rather than video memory.
- AGP is available in three different versions.: the original AGP version ,AGP 2.0,AGP 3.0 (AGP 8x)

- A computer with AGP support has one AGP slot next to all other expansion slots or an onboard AGP video.
- An Accelerated Graphics Port (AGP) is a point-to-point channel that is used for high-speed video output.
- This port is used to connect graphic cards to a computer's motherboard.
- It increases the speed at which machines can render graphics while using the system's resources more efficiently.
- The primary purpose of an AGP is to convey 3-D images much more smoothly than is possible on a regular PC.

Characteristics of AGP

- It has high quality and very fast performance.
- It has a direct path to the PC's main memory.
- It connects to the CPU and operates at the speed of the processor bus.
- It sends video information more quickly to the card for processing.
- It uses the main memory to hold 3D images.
- It provides the graphics card with two methods of directly accessing texture maps in system memory: pipelining and sideband addressing.
- The port is identified by its brown colour.

Advantages of AGP over PCI

- AGP was introduced as a replacement for the slower Peripheral Component Interconnect (PCI) interfaces.
- AGP provides a direct line of communication to the CPU and RAM, which in turn allows for quicker rendering of graphics.

AGP	PCI
Pipelined requests	Nonpipelined requests
Address/data demultiplexed	Address/data multiplexed
533 Mbps peak in 32-bit	133 Mbps peak in 32-bit
Single target, single master	Multitarget, multimaster
Memory read/write only; no other I/O	Linked to entire system
High/Low priority queues	No priority queues

Table 4-4. *Comparison of AGP and PCI System Bus*

PCI-X (Peripheral Component Interconnect Extended)

- PCI-X (Peripheral Component Interconnect Extended) is a computer [bus](#) technology (the "data pipes" between parts of a computer) that increases the speed that data can move within a computer from 66 [MHz](#) to 133 MHz.
- The technology was developed jointly by IBM, HP, and Compaq. PCI-X doubles the speed and amount of data exchanged between the computer [processor](#) and [peripherals](#).
- Up to 133 MHz bus speed.
- 64-Bit bandwidth.
- 1 GB/sec throughput.
- More efficient bus operation for easier interface.

On-board I/O devices

- At the back of the computer where the I/O connectors are located.
- There are three types of I/O connector layouts, also known as form factors.
- The two most common types are Baby-AT and ATX .
- Third type of form factor, called low profile.

Baby-AT

In the case of the Baby-AT systems, most connectors will be found on the adapter cards in their slots with only the keyboard connector coming directly off of the motherboard. In some cases, the serial and parallel port connectors may be mounted directly on the chassis instead of a card slot, but an internal cable is still used to connect the chassis to the adapter card or motherboard.

ATX

In the case of the ATX style, however, the keyboard, mouse, serial and parallel ports will almost always be mounted directly on the chassis and at the same time be soldered directly to the motherboard without cabling. These are called integrated or embedded ports or controllers. Most ATX systems today also have one or two USB ports and some may include the game connectors.

Low profile

In a low-profile design, the adapter cards do not plug directly into any motherboard slots. Instead, there is usually one vertically mounted riser card that rises up from the motherboard and holds the system I/O bus slots that the adapter cards will then plug into. This means the adapter cards will actually lie parallel to the motherboard and in case where only two or three slots are provided, this allows for a much shallower. These structures are often called pizza-box computers. All low-profile designs include sound, game and video connectors built in

and will often include a network interface as well. Older low-profile designs were proprietary to each manufacturer and not interchangeable, which usually made them much more expensive to replace.

I/O Interface Connections

- Keyboard port
- Floppy Controller
- Integrated Drive Electronics (IDE) Interface
- Serial and Parallel Communication Ports
- Audio, Video, Modem, Network and SCSI

Keyboard port

The PS/2 port is a 6-pin mini-DIN connector used for connecting keyboards and mice to a PC compatible computer system.

Floppy Controller

A floppy disk controller (FDC) is an electronic chip controller used as an interface between a computer and a floppy disk drive. Modern computers have this chip embedded in the motherboard. Earlier versions required a separate controller card that plugged into an ISA bus slot.

Integrated Drive Electronics (IDE) Interface

IDE (Integrated Drive Electronics) is a standard electronic interface used between a computer motherboard's data paths or bus and the computer's disk storage devices. Integrated Drive Electronics (IDE) is a standard interface for connecting a motherboard to storage devices such as hard drives and CD-ROM/DVD drives. The original IDE had a 16-bit interface that connected two devices to a single-ribbon cable. IDE is also known as Advanced Technology Attachment (ATA) or intelligent drive electronics (IDE) or PATA (parallel ATA). Each IDE channel can support two drives. A single IDE ribbon cable has three connection. One connection connects to the motherboard and the other two connections are available for the two drives.

Serial and Parallel Communication Ports

Initially most serial and parallel ports were available only on the adapter cards. Even though the serial and parallel ports may have been located on the motherboard, a cable was still required to connect from the *dual row header* connectors on the motherboard to the DB9 and DB25 external connectors mounted on the chassis. The connectors may be located on the fixed part of the chassis next to the power supply, or they may be mounted on slot covers and appear from the outside as if they are part of an adapter card. In ATX standard, which reserves space for the serial, parallel and other connectors to be soldered directly on the motherboard and also mounted on the chassis.

Audio, Video, Modem, Network and SCSI

These are interfaces that have been known to be mounted on the motherboard, either in nonstandard, ATX, or NLX systems. Usually, these other devices are best installed as adapter cards, so that they can be easily upgraded as advanced technology becomes available.

Chipsets

A chipset is a group of interdependent motherboard chips or integrated circuits that control the flow of data and instructions between the central processing unit (CPU) or microprocessor and external devices. A chipset controls external buses, memory cache and some peripherals. In addition, a CPU has several different chipsets that vary according to architecture. A chipset is specifically designed for a motherboard. The chipset and motherboard must be compatible with the CPU. A chipset has two sections – southbridge and northbridge – with specific sets of functions that communicate between the CPU and external devices. Modern motherboard contains a third chip is called Super I/O chip.

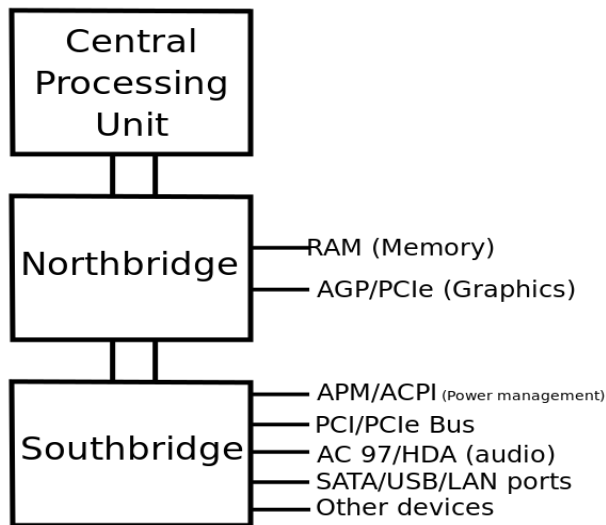
- Northbridge
- Southbridge
- Super I/O Chip

Northbridge

The northbridge is a chip inside a computer that connects the central processing unit (CPU) to other primary components in the system. These components include RAM (a.k.a. system memory), the frontside bus (FSB), PCI Express cards, and the AGP card. Unlike the southbridge, the northbridge is directly connected to these components. It acts as a "bridge" for the southbridge chip to communicate with the CPU, RAM, and graphics controller.

Southbridge

The southbridge is the component of the chipset that is responsible for interconnecting the CPU to the slower components of the system. The southbridge is a chip that connects the northbridge to other components inside the computer, including hard drives, network connections, USB and Firewire devices, the system clock, and standard PCI cards. The southbridge sends and receives data from the CPU through the northbridge chip, which is connected directly to the computer's processor. Since the southbridge is not connected directly to the CPU, it does not have to run as fast as the northbridge chip. the southbridge is also referred to as the I/O Controller Hub



Super I/O Chip

Technically speaking, the Super I/O chip is not part of the chipset. On most system boards, the Super I/O doesn't even come from the same manufacturer as the chipset. The chip controls the I/O functions that are typically found in all motherboards, including the serial ports, parallel ports, floppy drive controllers, IDE ports and so on. Also handled by the Super I/O on the majority of systems is the keyboard controller. Some Super I/O chips even take over the function of the real-time clock.

Breaking Tradition

Starting with the 820 chipset, Intel is beginning to move away from the traditional Northbridges/Southbridge design. A device called the memory controller hub has replaced the Northbridge component. This device, as its name infers, has control of all system memory. In addition, it assumes the responsibility for communicating with the AGP bus. The second chip, the enhanced I/O controller hub, handles all other functions.

Chipset manufactures

- Acer Laboratories, Inc. (ALI)
- Advanced Micro Devices (AMD)
- Eteq Components, Ltd.
- Intel Corporation
- OPTi
- PC Chips Manufacturing Limited
- Silicon Integrated Systems Corporation (SiS)
- VIA Technologies

Functions of the Chipset

Features of a computer that are specifically determined by the chipset include these:

- Processor voltage
- Number of processors
- Processor speeds supported
- Bus speed
- Types of memory supported
- Amount of memory supported
- Expansion buses supported
- Number of expansion slots addressed
- Power management
- Advanced device support

ROM BIOS

The BIOS is collection of very small programs, permanently stored on a ROM chip, which can be accessed from the computer on startup. These programs interpret the data coming from other devices and convert it into the commands the CPU can use. Short for Basic Input/Output System, the BIOS is a [ROM](#) chip found on [motherboards](#) that allows you to access and set up your computer system at the most basic level. The BIOS includes instructions on how to load basic computer hardware. It also includes a test referred to as a [POST](#) (Power-On Self-Test) that helps verify the computer meets requirements to boot up properly. If the computer does not pass the POST, you hear a combination of beeps indicating what is malfunctioning in the computer.

What are the four main functions of a PC BIOS?

- **POST** - Test the computer [hardware](#) and make sure no errors exist before loading the operating system. Additional information on the POST is available on our [POST and beep codes](#) page.
- **Bootstrap Loader** - Locate the [operating system](#). If a capable operating system is located, the BIOS will pass control to it. a bootstrap loader is a [program](#) that resides in the computer's [ROM](#) chip. It is automatically executed by the processor when turning on the computer. The bootstrap loader reads the [hard drives](#) boot sector to continue to load the computer's [operating system](#).
- **BIOS drivers** - Low-level drivers that give the computer basic operational control over your computer's hardware.
- **BIOS setup or CMOS setup** - Configuration program that allows you to configure hardware settings including system settings, such as computer passwords, time, and date.

ROM POST

A power-on self-test (POST) is a succession of built-in diagnostic tests performed when turning on a computer. When power is applied to the CPU, the first thing that it does is to execute the code located at memory address FFFF0(hex). This is the location in which all IBM-compatible computers store the first lines of the POST routine; the first step in what is known as the boot process. The first step of POST is to perform a system check to make sure all components are properly functioning. A Power On Self-Test checks that basic system devices are present and working properly, like the keyboard and other peripheral devices, and other hardware elements like the processor, storage devices, and memory.

The computer will continue to boot after the POST but only if it was successful. If the POST finds something wrong during its test, usually get an error of some kind. Errors might come in the form of flashing LEDs, audible beeps, or error messages on the monitor, all of which are technically referred to as POST codes, beep codes, and on-screen POST error messages, respectively.

Types of ROMs and their characteristics

- MROM (Masked ROM)
- PROM (Programmable Read Only Memory)
- EPROM (Erasable and Programmable Read Only Memory)
- EEPROM (Electrically Erasable and Programmable Read Only Memory)

MROM (Masked ROM)

The very first ROMs were hard-wired devices that contained a pre-programmed set of data or instructions. These kinds of ROMs are known as masked ROMs, which are inexpensive.

PROM (Programmable Read Only Memory)

PROM is read-only memory that can be modified only once by a user. The user buys a blank PROM and enters the desired contents using a PROM program. Inside the PROM chip, there are small fuses which are burnt open during programming. It can be programmed only once and is not erasable.

EPROM (Erasable and Programmable Read Only Memory)

EPROM can be erased by exposing it to ultra-violet light for a duration of up to 40 minutes. Usually, an EPROM eraser achieves this function. During programming, an electrical charge is trapped in an insulated gate region. The charge is retained for more than 10 years because the charge has no leakage path. For erasing this charge, ultra-violet light is passed through a quartz crystal window (lid). This exposure to ultra-violet light dissipates the charge. During normal use, the quartz lid is sealed with a sticker.

EEPROM (Electrically Erasable and Programmable Read Only Memory)

EEPROM is programmed and erased electrically. It can be erased and reprogrammed about ten thousand times. Both erasing and programming take about 4 to 10 ms (millisecond). In EEPROM, any location can be selectively erased and programmed. EEPROMs can be erased one byte at a time, rather than erasing the entire chip. Hence, the process of reprogramming is flexible but slow.

Advantages of ROM

- Non-volatile in nature
- Cannot be accidentally changed
- Cheaper than RAMs
- Easy to test
- More reliable than RAMs
- Static and do not require refreshing
- Contents are always known and can be verified

CMOS setup

Unlike the BIOS, which cannot be edited or reconfigured by the user, CMOS is where certain parameters that relate to BIOS can be changed. When a computer is booting up, the BIOS checks the CMOS for its settings and uses those settings in conjunction with the routines that it executes. The CMOS stores parameters for setting up devices that every system uses, but that may change over the life of the system. Once entered by the user, the computer retains CMOS information even after it is turned off, providing that the onboard battery retains its charge. By taking away the battery charge, CMOS settings are lost.

Settings under the control of the CMOS setup include the following:

- Basic settings, which include real-time clock settings, floppy disk drive settings, and on some brands of BIOS, the boot sequence of bootable devices.
- Hard disk drive detection and configuration.
- Configuration of PnP and PCI devices. This may include reserving IRQs for use by ISA.
- Security settings, including user and administrator passwords and in some cases the ability to disable certain devices such as floppy drives.
- Power management settings.

CMOS Settings

- Every computer provides a way to enter the [BIOS](#) or [CMOS](#) setup. This interface allows you to configure the basic settings of your computer.

- Computers manufactured in the last few years allow you to enter the BIOS setup using one of the five keys shown below during the [boot](#) process.
- F1
- F2 *
- F10 **
- [Del](#)
- [Esc](#)

The CMOS setup, often accessed through the BIOS (Basic Input/Output System), allows users to configure various settings related to the computer's hardware. The specific options can vary between different BIOS versions and motherboard manufacturers, but here are some common basic options you might find in the CMOS setup:

1. **Date and Time:**
Allows you to set the system date and time.
2. **Boot Order:**
Specifies the order in which the computer looks for an operating system to load. You can prioritize booting from different devices such as hard drives, optical drives, USB drives, etc.
3. **System Information:**
Displays basic information about your computer, including the BIOS version, CPU type, and installed memory.
4. **Password Settings:**
Lets you set or modify passwords for accessing the BIOS or booting the system. This can include supervisor passwords and user passwords.
5. **Advanced Chipset Options:**
Provides access to advanced settings related to the motherboard chipset, such as enabling/disabling certain features or adjusting memory timings.
6. **Integrated Peripherals:**
Configures settings for onboard peripherals like USB controllers, SATA controllers, audio, and network interfaces.
7. **Power Management:**
Controls power-related settings, including sleep modes, wake-on-LAN, and power-saving features.
8. **CPU Configuration:**
Allows for adjustments to CPU-related settings, such as clock speed, multiplier, and sometimes voltage settings (overclocking features).
9. **Security Options:**
Manages security-related settings, including options for trusted computing and hardware-level security features.

10. **Virtualization Support:**
Enables or disables hardware virtualization features if supported by the CPU.
11. **Hardware Monitoring:**
Monitors system temperatures, voltages, and fan speeds. Some setups allow for adjusting fan speed profiles.
12. **Boot Device Priority:**
Similar to boot order, this allows you to specify the priority of specific devices during the boot process.
13. **Exit Options:**
Provides options to save or discard changes made in the CMOS setup and exit the BIOS.

Motherboard physical form factors

The form factor is the physical layout of the motherboard. This determines the size and orientation of the board in relation to the case, determines the type of power supply required and dictates what onboard peripherals might be located on the board.

- AT Motherboards
- ATX Motherboards
- LPX Motherboards
- NLX Motherboards

AT Motherboards:

Form Factor: Advanced Technology (AT) motherboards were the standard in the 1980s and early 1990s.

Size: Larger and more rectangular compared to modern standards.

Power Supply: The power supply was typically located on the top rear of the case.

Peripheral Connectors: Featured serial and parallel ports for peripherals.

Expansion Slots: Included ISA (Industry Standard Architecture) slots for expansion cards.

Legacy Standard: Phased out in favor of smaller and more efficient form factors.

ATX Motherboards:

Form Factor: Introduced by Intel in 1995 as an evolution of the AT design.

Size: Rectangular shape with a standardized layout.

Power Supply: Power supply is typically integrated into the case and positioned at the top rear.

Peripheral Connectors: USB, audio, and other ports were integrated directly onto the motherboard.

Expansion Slots: PCI (Peripheral Component Interconnect) and later PCIe (PCI Express) slots replaced ISA slots.

Improved Airflow: Designed for improved cooling with the CPU and memory positioned to allow better airflow.

Backward Compatibility: ATX is backward compatible with AT power supplies and cases.

LPX Motherboards:

Form Factor: Low-Profile Extended (LPX) motherboards were developed by Western Digital.

Size: Smaller and more compact compared to AT and ATX.

Orientation: CPU and memory sockets are at a right angle to the expansion slots, allowing for a more compact design.

Power Supply: Power supply is often mounted at the front of the case.

Peripheral Connectors: Features integrated ports similar to ATX.

Legacy Standard: Has become obsolete as ATX and other form factors gained popularity.

NLX Motherboards:

Form Factor: New Low Profile Extended (NLX) motherboards were introduced to address the limitations of LPX.

Size: Compact and designed for easy assembly and upgrades.

Orientation: Similar to LPX, with right-angled CPU and memory sockets.

Power Supply: Uses an external power supply to further reduce the size of the case.

Peripheral Connectors: Integrated ports for peripherals.

Expansion Slots: Uses riser cards for additional expansion slots.

Legacy Standard: Became less common as ATX became the dominant standard for compact systems.

Power supply

The PC power supply is primarily designed simply to take AC power from the wall outlet and convert it into the much lower DC voltage needed by the system. It must also do this conversion reliably and efficiently and must be able to shut down the system if it is overloaded, short-circuits, or overheats. When the PC power supply starts up, it must do so in the proper sequence. When it shuts down, it must prevent any voltage overshoot or fluctuations so that none of the sensitive circuits are exposed to any damaging voltage fluctuations. It must test itself continuously, so that any detected failure will cause an immediate and safe shutdown.

Power Supply Function and Operation

- Converting AC to DC Power
- Output Supply Voltage Lines
- Power Supply Control Signals

How to test computer SMPS

To check the supply, whether it is working properly or not, you can test it before fitting it in the cabinet. There is a simple way to test the SMPS, you just have to short the green wired connector to any black colored(ground) connector. If the SMPS is working perfectly, the fan of the supply will start working.

Power Supply Testing Equipment

- DC voltmeter or multimeter
- Logic Probe
- Dummy power load
- Variable transformer
- True-RMS voltmeter
- Oscilloscope

Power Supply Form Factors

A **form factor** is defined as a **size**, **shape**, and set of **features**. Some types of system board designs are listed: AT, Baby AT, ATX, LPX, and NLX Each has variations.

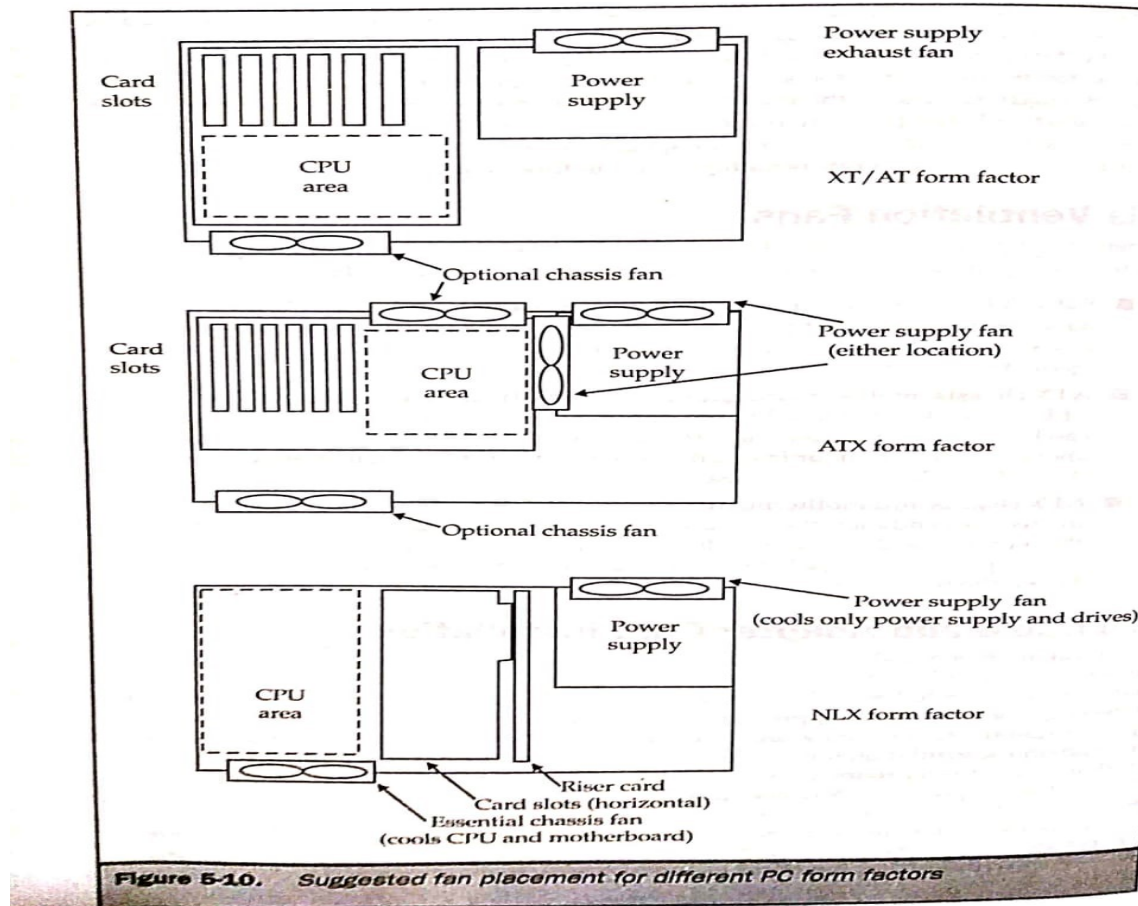
1. **AT (Advanced Technology):**
Description: The original power supply form factor used in early personal computers.
Size: Larger and boxy.
Connectors: AT power connectors for the motherboard.
2. **ATX (Advanced Technology eXtended):**
Description: Introduced by Intel in 1995, it became the standard for desktop computers.
Size: Rectangular shape with a fan on top.
Connectors: ATX power connectors, providing additional features like soft power on/off and standby power.
Compatibility: Backward compatible with AT.
Common Versions: ATX12V, EPS12V.
3. **MicroATX:**
Description: A smaller version of ATX, designed for smaller form factor motherboards.
Size: Reduced dimensions while maintaining ATX connector compatibility.
Common in: Compact desktops and smaller computer cases.

Ventilation and cooling protection

- Power Supply Fan : Airflow Direction
- Power Supply Fan Placement

Chassis Ventilation Fans

- Baby-AT Chassis design
- ATX chassis, motherboard and power supply design
- NLX chassis and motherboard design



Baby-AT Chassis design

Most Baby-AT designs have room on the front panel for a chassis cooling fan. There is no guarantee, however, that the CPU will be anywhere near this fan, so the CPU should have its own cooling fan as well.

ATX chassis, motherboard and power supply design

In the ATX design, the CPU is specifically placed in the right-rear corner of the motherboard, so that it can benefit directly from the airflow going through the power-supply fan either above it or next to it, or from an auxiliary chassis fan installed directly above the CPU on the rear of the case.

NLX chassis and motherboard design

Part of the NLX design specification is the recommendation that the processor be located near the front left corner of the motherboard, so that it will not interfere with the adapter cards on the right or I/O ports on the back panel, and so that it can be cooled directly by a chassis fan on the front panel.

Ribbon Cable and Adapter Card Installation

The installation of any flat ribbon cables must be done carefully, because improper cable routing can block airflow or cause short circuits. Always make sure to use a cable that's long enough to have a little slack in it, especially if it will need to go around one of the edges of the power supply. If the cable can barely reach and is stretched fairly tight, vibration over time may cause the cable to become disconnected, or worse-.

Generally, you want to put any adapter cards with ribbon cables as close as possible to the drives they connect to, unless they are full-length cards in a Baby-AT case. In such a scenario, the full-length card may block the airflow to the CPU, and so it should not be placed between the CPU and any ventilation fan, if possible.

Processor Cooling

- CPU heat sink
- CPU cooling fan
- Chassis or power supply fan
- Peltier-Junction cooling elements

Power protection and backup

Using high quality uninterruptible power supply (UPS) equipment can help protect your hardware against damage from unexpected power failures or unclean power. UPS devices provide power backup to your system, ensuring that there is no interruption between your computer system and the main power supply. Even the best power supply is helpless without a source of reliable, steady AC power. Simply plugging your system into a wall receptacle and hoping for the best is a sure road to disaster, sooner or later. Before we smartened up, we lost many hours' work to power failures, and more than one system to lightning damage. All of that was preventable, if only we'd installed proper power protection.

Some power quality disturbances are

1. Transients
2. Interruptions
3. Sag / Undervoltage

4. Swell / Overvoltage
5. Voltage fluctuations
6. Frequency variations

Transients: Sudden, short-lived voltage spikes or surges that can potentially damage electronic equipment.

Interruptions: Complete loss of power for a brief duration, leading to a temporary shutdown of electronic devices.

Sag/Undervoltage: Temporary reduction in voltage levels below the standard, which may cause equipment malfunction or damage.

Swell/Overvoltage: Temporary increase in voltage levels above the standard, posing a risk of damage to connected devices.

Voltage Fluctuations: Variations in voltage levels, including both sags and swells, which can affect the performance and lifespan of electronic equipment.

Frequency Variations: Changes in the frequency of the electrical power supply, which can impact the operation of devices dependent on specific frequency requirements.

Notching

Notching is a periodic voltage disturbance caused by electronic devices,

Noise

Noise is unwanted voltage or current superimposed on the power system voltage or current waveform.

There are many different approaches to controlling noise

- Isolate the load via a UPS
- Install a grounded, shielded isolation transformer
- Relocate the load away from the interference source
- Install noise filters
- Cable shielding

Backup Power Systems

A **backup power system** is used to provide energy when the primary source fails. This **system** is very important since an **uninterruptible power** supply is crucial for any operation. An uninterruptible power supply or uninterruptible power source (**UPS**) is an electrical apparatus that provides emergency power to a load when the input power source or mains power fails.

UPS System Types

- SPS (Standby Power Source)

- True-Online UPS
- Ferro-resonant UPS
- Hybrid-online UPS
- Line-interactive UPS

SPS (Standby Power Source)

An **SPS** contains a **battery** like the UPS, but the **battery** provides **power** to the computer only when it loses AC **power**. It does not provide constant **power**, like the UPS. Short for **standby power supply**, an **SPS** contains a circuit that is capable of telling if and when a disturbance in electricity occurs. When a disturbance occurs, the SPS switches to battery power.

True-Online UPS

Under normal operation, a **true online UPS** runs continually on the battery via the inverter, while the line power runs the battery charger.

Ferro-resonant UPS

A **ferroresonant**-based **UPS** uses an internal line-conditioning transformer to provide voltage regulation, noise and harmonics filtering, spike suppression, and galvanic isolation.

Hybrid-online UPS

Standby **online hybrid UPS**, also referred to as double conversion on demand, are similar to true **online UPS**, but with higher energy efficiency. ... Once the power exceeds the set tolerances, the **UPS** switches to standard true **online** operation.

Line-interactive UPS

Line-interactive UPS systems use automatic voltage regulation (AVR) to correct abnormal voltages without switching to battery. The **UPS** detects when voltage crosses a preset low or high threshold value and uses transformers to boost or lower the voltage by a set amount to return it to the acceptable range.